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Narcolepsy in Parkinson's disease

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Non-motor symptoms in Parkinson's disease (PD), such as excessive daytime sleepiness, 'sleep attacks', insomnia, restless legs syndrome and rapid eye movement sleep behavior disorder, are common and provide a challenge to treatment. These sleep symptoms are also described in patients suffering from the sleep/wake disorder, narcolepsy. The International Classification of Sleep Disorders (ICSD-2) narcolepsy criteria uses a number of markers for diagnosis, of which lack or deficiency of cerebrospinal fluid (CSF) hypocretin-1 levels is a key marker. Hypocretin neurons prominently located in the lateral hypothalamus and perifornical nucleus have been proposed to interact with mechanisms involving sleep and arousal. Low hypocretin-1 levels in the CSF have been shown to correlate with hypothalamic hypocretin cell loss in narcolepsy and other forms of hypersomnia; therefore, it has been proposed that degenerative damage to hypocretin neurons (such as in PD) may be detected by low CSF hypocretin-1 concentrations, and may also explain the sleep symptoms experienced by some PD patients. To date, there is mixed conflicting data describing hypocretin-1 levels in the CSF of patients with parkinsonism associated with sleep symptoms, with most studies showing no significant decrease when compared with controls. However, hypocretin-1 CSF deficiency has been shown in some studies to be more prominent in PD patients with sleep symptoms versus those without. Notably, the hypocretin system has been shown not to be selectively disrupted, with one study showing melanin concentrating hormone cell loss in the same patients with hypocretin loss. It is likely that hypocretin deficiency in PD patients occurs secondary to collateral damage caused by the neurodegenerative process involving the hypothalamus. Awareness of narcoleptic events in PD is important for driving related advice, in addition to the possible use of dopamine D3 receptor active agonists.

KEYWORDS: EDS • excessive daytime sleepiness • hypocretin • narcolepsy • non-motor symptoms • orexin • Parkinson's disease • rapid eye movement sleep behavior disorder • RBD • REM

Introduction & etiologies

Narcolepsy

Narcolepsy is a disorder characterized by dysfunction of the systems associated with sleep and waking from sleep. It has been shown to affect approximately 1:2000 people in Europe [101]. Three million people worldwide are thought to be sufferers, although the reported number is much lower as a result of frequent under-diagnosis. The condition is thought to start in adolescence, with the highest frequencies of sufferers aged 15–30 years [1,2]. Sleep symptoms involve excessive daytime sleepiness (EDS), disturbed nocturnal sleep, hallucinations and rapid eye movement (REM) sleep abnormalities. Other than the prominent sleep symptoms, patients may have associated cataplexy – a loss of muscle tone elicited by strong emotion. It is prevalent in approximately 75% of narcoleptics and is often used to categorize the type of narcolepsy

(i.e., with or without cataplexy). Emotions that elicit these episodes are intense, commonly laughter, anger and amusement [3]. The resulting loss of tone can cause the collapse of facial muscles, lower limb joints and slurring of speech. Increased BMI is noted in narcoleptics, due to interactions between the systems involving sleep and energy balance [4–9].

Parkinson's disease

Parkinson's disease (PD) is a chronic and progressive disorder characterized by resting tremor, rigidity, postural instability and, most notably, bradykinesia. Degeneration of dopaminergic neurons in the substantia nigra pars compacta results in decreased dopamine signaling from the basal ganglia to the motor cortex, leading to clinical motor complications in PD. Other motor symptoms include gait disturbance, dysarthria, dysphagia and micrographia.

Non-motor symptoms of PD

The non-motor symptoms of PD are key determinants of quality of life, and sleep symptoms, such as EDS, are common [10]. Moreover, non-motor symptoms have been shown to cause more institutionalization and a lower quality of life than the characteristic motor symptoms of PD that are treatable with dopamine stimulation [11,12]. Non-motor symptoms tend to present themselves more frequently as the disease progresses, although exceptions include impaired olfaction, constipation, depression, EDS and REM sleep behavior disorder (RBD) [13]. A possible explanation for this can be derived from the 'bottom-up' pathogenesis theory of Braak and colleagues, who proposed that dopaminergic transmission loss was consistent with Lewy body distribution in PD and six stages of symptom emergence were identified [14]. Braak stage 1 involves Lewy body formation at the olfactory bulb [15]. This explains the early olfactory symptoms experienced by PD patients. Stage 2 notes lesions in the inferior brainstem, accounting for the autonomic and sleep disturbances, such as EDS and RBD, which may be in part mediated via the locus ceruleus and raphe nucleus. However, it is important to note that the Braak hypothesis uses Lewy body dispersal to explain loss of function, rather than actual dopamine neuron degeneration.

Sleep symptoms in narcolepsy & PD

Sleep in narcolepsy

The sleep symptoms in narcolepsy are complex, with the primary and most prominent being EDS. In EDS, sleep attacks masquerade as patients having an uncontrollable tendency to inappropriately fall asleep during daytime hours. This can often lead to social denunciation, as the patient's actions may be deemed to be offensive or portraying a lack of interest by others. To evaluate EDS, the Epworth Sleepiness Scale is often used (>10 out of 24), with a measurement of sleep latency using the Multiple Sleep Latency Test. Additionally, narcoleptics experience disturbed nocturnal sleep that may accentuate episodes of EDS. These are often characterized by a heightened sense of alertness, tachycardia or periodic limb movements. Hypnagogic and hypnopompic hallucinations are also present and can be auditory, color visual or tactile in nature. Involuntary sleep paralysis also occurs in narcolepsy, where the patient is unable to move from a fixed position, often during bedtime or upon arousal from sleep. Moreover, patients may present with automatism, where routine tasks are performed by the patient but not under conscious control [16,17]. There are also complications involving the REM stage of sleep in narcoleptics. REM sleep normally occurs approximately 1 h into sleep, although with narcoleptics this seems to occur much sooner, often within minutes [102]. RBD is a characteristic feature, where patients have involuntary limb movements and often act out violent dreams [18].

Sleep in PD

There are evident similarities in the sleep symptoms experienced by PD patients and those diagnosed with narcolepsy [19]. Sleep disturbances occur very regularly in PD and some appear early (RBD and EDS have both been identified as premotor features

consistent with Braak stage 2) in the course of the disease [20,21]. As in narcolepsy, EDS is also apparent in PD patients [22]. This may be due to degeneration of sleep pathways and disrupted nocturnal sleep due to frequent arousal, and dopaminergic treatment is also implicated [23–25]. During RBD, sufferers can kick, jump out of bed and act out violent dreams, which may be distressing to the patient and bed partners, with severe cases reporting inadvertent homicide [26]. These symptoms have been described in PD to be due to a lack of skeletal muscle atonia that normally occurs during REM sleep [27]. Involvement of the pedunculopontine and the sublaterodorsal nucleus as the potential site of RBD pathology has been discussed [28]. In addition, hallucinations and REM disturbances in PD have also been documented [29–31].

Hypocretin/orexin neurotransmission

Background & development

The sequences for the genes encoding orexin/hypocretin, as well as localizing the neurons that produce the neuropeptide, were identified concurrently by two separate groups [32,33]. Sakurai and colleagues were able to identify two distinct hypocretin proteins, hypocretin-1 and -2 (cleaved from their common precursor, prepro-hypocretin), and their receptors, Hcrtr1 (OX₁R) and Hcrtr2 (OX₂R) [34]. Neuronal peptides were shown to be concentrated in the lateral hypothalamus of the rat and the dorsomedial hypothalamic nucleus [32]. Gautvik's group later identified these peptides in the lateral hypothalamus to be hypocretins, the name derived from its location (hypothalamus) and the structure being similar to the duodenal hormone, secretin [33]. Hypocretin-1 was shown to have equal affinity for both receptors, while hypocretin-2 binds preferentially to the Hcrtr2 receptor. Although mainly concentrated around the hypothalamus, hypocretins have been shown to project throughout the brain [35].

Hypocretin/orexin neurophysiology in sleep

The hypocretin neurotransmission system was shown to be disrupted in narcoleptic canines through an OX₂R mutation by Lin and colleagues [36]. A Texan group, through electroencephalographic and behavioral data of mice, observed that that orexin^{-/-} mice exhibited a similarity in phenotype to human and canine narcoleptics, with cataplexy and activation of orexin-containing neurons by the antinarcotic drug modafinil being pivotal findings [37,38]. Other than the lateral hypothalamus, the hypocretin system has neurons extending throughout monoaminergic cell groups. The proposed method by which hypocretin mediates sleep arousal may involve its actions on serotonergic, noradrenaline (catecholaminergic) and histaminergic pathways [35,38–45]. Saper and colleagues proposed a 'flip-flop' switch control for wakefulness. This is theorized to involve the suprachiasmatic nucleus and hypocretin [46]. The latter is thought to control three monoaminergic neuronal centers and pathways: catecholaminergic in the locus ceruleus; histaminergic in the tuberomammillary nucleus; and serotonergic in the raphe nuclei. It is proposed that hypocretin stimulates arousal through the activation of these pathways and centers [47].

Preclinical markers & CSF hypocretin

Preclinical markers for PD

A major therapeutic challenge in a neurodegenerative disease such as PD would be to identify and subsequently intervene, should neuroprotection be available, early in the disease process. A number of studies have examined associated non-motor PD symptoms in the premorbid state as potential markers for eventual disease development. Longitudinal data from Schenck and colleagues examined 29 patients diagnosed with idiopathic RBD and found that approximately 38% of patients subsequently went on to develop parkinsonism [28]. There have also been reported associations between patients suffering from olfactory disturbances and constipation, and the subsequent development of PD [48,49]. EDS is also implicated, and Abbott and colleagues conducted a 10-year study of over 3000 men, aged 71–93 years, with reported EDS but no PD or dementia [50]. Follow-up of these patients with neurological examinations in subsequent years found the development of PD to be three-times more prevalent in patients with EDS than those without. Iranzo and colleagues conducted a retrospective study of 44 patients diagnosed with idiopathic RBD [51]. They found, subsequent to full neurological examination and testing, that 20 out of the 44 patients developed a neurological disease within a mean of 11.5 years from the reported RBD onset. The similarity of sleep symptoms noted in PD and narcolepsy suggest that it is likely that the mechanisms underlying these symptoms may also be similar [50].

Hypocretin CSF levels in narcolepsy

An interesting discovery in 2000 by Nishino and colleagues found low levels of cerebrospinal fluid (CSF) hypocretin in human narcoleptics [52]. The group found that seven out of nine narcoleptic patients had hypocretin-1 levels of less than 40 pg/ml, versus none less than 250 pg/ml in eight controls. Further to this, Mignot and colleagues established that low CSF hypocretin levels are a definitive diagnostic test for narcolepsy (<110 pg/ml), especially in cases where associated symptoms of narcolepsy (e.g., cataplexy) cannot be identified [53]. The sleep-promoting effects of the hypocretin system were demonstrated further by Espana and colleagues, where hypocretin-1 infused into the lateral ventricles of animals caused a longer period of waking without sleep [39]. It is known that the majority of narcoleptic disorders are nonfamilial. Nishino and colleagues analyzed 116 narcoleptic cases in literature that met the International Classification of Sleep Disorders (ICSD) criteria. They reported that 33% of cases were due to inherited disorders and 44% of cases were due to organic, nongenetic causes (tumors = 28% and head trauma = 16%) [54]. Subsequently, 9 months after the findings of the Nishino *et al.* study, Thannickal and colleagues showed that there is a selective 85–95% reduction in hypocretin neurons from the hypothalamus in human narcoleptics [55]. They further identified that gliosis in the hypothalamus implied neuronal degeneration was consistent with the hypocretin cell loss. Normal levels of CSF hypocretin-1 have been recorded as an average of 350 pg/ml [54], although they are thought to be significantly higher in brain tissue at between 500 and 1000 pg/ml [56,57]. A correlation between actual hypocretin neuronal function and CSF levels of hypocretin-1 has been demonstrated by Mignot and colleagues [53] and in rodents

by Gerashchenko and colleagues [58]. They found that administration of the hypocretin neurotoxin HCRT2-SAP in the lateral hypothalamus led to a 50% decline in CSF hypocretin-1 by day 60. Moreover, this was shown further in humans by Fronczek and colleagues [57]. This implies that hypocretin CSF levels may be diagnostically consistent with hypocretin cell loss. The ICSD-2 diagnostic criteria for narcolepsy now documents CSF hypocretin as a marker for narcolepsy [59].

Hypocretin CSF levels in PD

After the discovery of low CSF hypocretin-1 in narcoleptics, similar work was conducted on a range of neurological conditions that exhibited narcoleptic-like sleep disturbances [60]. One such study by a group in the USA examined CSF hypocretin-1 levels in narcoleptic, PD, Alzheimer's disease, multiple sclerosis and Guillain-Barré syndrome patients. They found undetectably low levels of hypocretin-1 in 90% of narcoleptics; however, other than three Guillain-Barré syndrome patients (undetectable) and a handful of patients with trauma or tumors (low hypocretin [100–194 pg/ml]), those with neurological conditions (including those with PD) exhibited CSF levels that fell within the normal range of control values [61]. In a subsequent study, Overeem and colleagues selected PD patients that specifically had sleep disturbances (EDS) for CSF hypocretin analysis. However, the group did not find a significant hypocretin concentration difference between PD patients and controls [62]. In 2003, a French group collected ventricular CSF (as opposed to lumbar CSF used in other studies) from 19 patients with advanced PD. They reported significantly reduced orexin levels in patients with PD (57 pg/ml) compared with controls (199 pg/ml). Moreover, they found undetectable levels of orexin in nine of the 19 patients [63]. Interestingly, Drouot's team also found that there was a correlation between lower CSF orexin levels and advancing disease state. Two papers by Baumann *et al.*, in 2004 and 2005, analyzed CSF hypocretin in patients with EDS associated with PD and EDS patients with dementia with Lewy bodies (DLB) [60]. In the DLB study, all ten patients had hypocretin-1 levels similar to controls (DLB mean: 521 pg/ml vs control mean: 497 pg/ml). The EDS in PD study reported only two patients with low CSF hypocretin (221 and 307 pg/ml), compared with eight in the normal range, and all were comparable to controls (mean: 497 pg/ml, as previous). As in the Mignot study, Baumann and colleagues selected patients with EDS and sleep disturbances, and hence showed abnormal sleep symptoms in PD, with CSF hypocretin-1 intact [60]. In 2006, Yasui and colleagues [64] and Maeda and colleagues [65] reported contrary evidence. The former was a relatively large study looking at 62 patients with EDS and PD, which found 61 of them to be within the normal CSF orexin range (>200 pg/ml) and only one in the high-intermediate range (183 pg/ml). Yasui *et al.* also analyzed EDS patients with DLB, progressive supranuclear palsy and corticobasal degeneration. They found that low orexin levels were present in patients with corticobasal degeneration and progressive supranuclear palsy, although only in the latter did low orexin levels correlate with disease duration. They found DLB and PD patients predominantly maintained CSF orexin concentration. Conversely, Maeda and colleagues reported a single case study of a 58-year-old

man with PD (for 15 years) and diagnosed with EDS. They showed that his CSF hypocretin levels were low at 86 pg/ml (no control). Recent studies are still conflicting, with normal [66] and low [67,68] levels of CSF hypocretin-1 in patients with sleep disturbances in PD being reported. Compta and colleagues considered three groups of patients: 21 PD patients with EDS and dementia; 20 PD patients with EDS without dementia; and 22 controls with no EDS [66]. They found that there was no significant difference between any of the groups with respect to CSF hypocretin-1 levels (309.94 ± 65.95 , 300.99 ± 58.68 and 321.15 ± 47.15 pg/ml, respectively). In fact, only two out of 42 PD patients had a distinctly low CSF hypocretin-1 level, and even those (115.93 and 196.75 pg/ml) were higher than the proposed upper limit for narcoleptics (<110 pg/ml).

Hypocretin neurons in brain tissue

Ratios between the hypocretin-1 level in the CSF and hypocretin neurons in the brain have been discussed [53]. Two significant studies in 2007 analyzed cadaveric PD brains, attempting to quantify the extent of hypocretin cell loss in brain tissue. Fronczek and colleagues evaluated hypocretin levels in three brain compartments. Firstly, the group analyzed post-mortem ventricular CSF levels of hypocretin in PD patients versus controls and found there to be a 20% reduction of hypocretin-1 in PD patients (PD mean: 365.5 pg/ml vs control mean: 483.5 pg/ml) [57]. They also analyzed hypocretin-1 concentrations in the cerebral cortex (prefrontal cortex) of PD patients versus controls and again found a decrease, this time of over 40%, in PD brains (PD mean: 389.6 pg/ml vs control mean: 676.6 pg/ml). Finally, they report an estimation of hypocretin cells via immunohistochemistry on nine PD hypothalami and matched controls. Their results showed that PD brains had 45% less hypocretin neurons (PD mean: 20,276 vs control mean: 36,842). All their results point towards a significant loss of hypocretin, in brain tissue and CSF. Thannickal and colleagues performed a similar study, where they immunostained cadaveric hypothalami, again in an effort to quantify hypocretin cell loss [69]. Using Hoehn and Yahr PD staging criteria, they found that hypocretin cell count decreased between normal sections (129), PD stage III (106) and PD stage V (72). Unlike the Fronczek study, they were able to show significantly decreasing hypocretin densities with disease stage progression (not disease duration). However, Thannickal and colleagues also showed that melanin concentrating hormone cells decreased with disease duration, which implies that the destruction of cells in the hypothalamus by PD is not selective to hypocretin neurons.

Discussion

Therapeutic potential

The similarities noted between the sleep symptoms of narcolepsy and those experienced by parkinsonian patients led to the hypothesis that the two diseases may have the same underlying dysfunction. As many of the sleep symptoms in PD occur before the onset of disease, identifying a cause for the disabling sleep symptoms in PD could lead to the development of a marker that evaluates risk of PD onset.

There is also clinical experience that some patients appear to be particularly sensitive to 'sleep attacks' while being started on a dopaminergic drug, particularly agonists with potent effects on D3

receptors. It remains to be confirmed whether such effects are manifest in a subset of PD patients who are hypocretin deficient. Such an effect appears particularly evident during the 'titration-up' phase of agonists and may have important implications on driving when patients are being initiated on certain dopamine agonists [70]. Use of modafinil may be of particular benefit to these patients, and there is some data suggesting that avoiding D3 agonists may be useful.[70]

Inconsistent results

In light of this, several studies have analyzed levels of CSF hypocretin-1 in many categories of PD and parkinsonism patients, such as those who are advanced in disease [63], those selected for severe sleep episodes [22,53], as well as post-mortem material [57,69]. Results have been conflicting. Post-mortem studies by Thannickal [69] and Fronczek [57] together showed lower levels of hypocretin-1 in ventricular CSF, less hypocretin neurons in the cortex and in hypothalamic tissue of patients suffering from PD versus controls. Moreover, the Thannickal study showed hypocretin neuronal loss advancing with PD disease progression (as per Hoehn and Yahr staging). Conversely, many studies conclusively report no significant decrease in the CSF hypocretin-1 levels of PD patients [61,62,64,66]. This may, in fact, be expected, as most cases of low CSF hypocretin-1 occur in narcoleptics with cataplexy than without, with the latter being absent in PD. Additionally, Thannickal and colleagues showed degeneration of other cell types (melanin-concentrating hormone) alongside hypocretin cells, which implies that the sleep symptoms may not be due to selective loss of hypocretin, but rather, diffuse cell damage due to PD disease progression [70].

Expert commentary & five-year view

Undoubtedly, obtaining preclinical markers for a chronic, progressive neurodegenerative disease such as PD would provide a multitude of interventional opportunities. This would change current therapeutic aims from managing or curing PD to developing preventative and neuroprotective pharmacological treatment. With respect to CSF hypocretin-1, it is important to note that more conclusive evidence is needed to establish whether there is any use for it as a potential prognostic marker. If future studies show specific hypocretin-1 degeneration to be correlative to sleep symptoms in PD, then research should aim at establishing the exact correlation between CSF hypocretin-1 and actual hypocretin functionality in sleep, and developing hypocretin analogues/agonists to establish (any) benefit or reversibility in sleep symptoms. However, as current data do not support the above, it is important to identify other plausible explanations for the evident sleep disturbances in PD, with the view that the hypocretin system is being involved incidentally due to the widespread degeneration in PD.

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Key issues

- In total, 20–30% of parkinsonian patients may have susceptibility to sudden-onset sleep or 'sleep attacks' that, in part, may resemble narcolepsy, although cataplexy is not a feature of this variant.
- Such patients may be particularly susceptible to the sedating effect of dopaminergic drugs, in particular dopamine agonists with robust action at D3 receptors, most notably during the 'titrating-up' stage of agonists.
- In some, deficiency of hypocretin, as part of overall hypothalamic damage as seen in Parkinson's disease, may be responsible.
- Alertness-promoting agents, such as modafinil, may be useful and further research is needed to refine and define this phenotype.

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